



UNIVERSITY OF GENEVA

Faculty of Science

Video Compression and Modification Detection

Digital Forensics

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Michel Jean Joseph Donnet

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1 Introduction

Nowadays, a large number of videos circulate on the internet. Given that each video is composed of around twenty images per second, storing and transmitting videos requires a large amount of data. This is why methods have been implemented to reduce the amount of data used by a video while preserving its visual quality. However, some videos circulating on the internet convey a distorted image of reality through clever modifications to the original content, which can even lead to people being exonerated in court. This is why it is important to know the history of a video, which can be achieved through digital forensics. Since video compression leaves traces, these can be analyzed to reveal any modifications to the video. The objective of this work will therefore be to understand how video compression works and to see how the traces left by compression can be used to detect modifications to the video.



2 Methodology

In general, video are sequence of images called frames. Based on this kind of video, we will start by studying how video compression works. To do this, we will use the h264 codec, as it is the most widely used codec on the web.

2.1 Redundancies

2.1.1 Psychovisual redundancies

The imperfections of the human visual system can be exploited to reduce the amount of data used by a video without a loss of visual quality.

2.1.1.1 Spatial irrelevance

The human visual system has some difficulties to perceive small details due to its limitations. The optics of the eyes and the neural processing tend to smooth out fine patterns. For example: a document printed by a laser printer is composed of small dots that are very close together. However, we cannot see the individual dots at a normal distance: we only perceive a uniform surface.

This is why this feature can be used to remove small details that are invisible or barely visible for human visual system.

2.1.1.2 Spectral irrelevance

The human visual system consists of approximately 100 million rods responsible for perceiving brightness and approximately 6.5 million cones responsible for perceiving colors. Due to the high amount of rods, the human visual system is more sensitive to brightness compared to color.

There are 3 types of cones:

- L-cones ($\approx 65\%$), sensitive to long wavelengths (such as the red color).
- M-cones ($\approx 33\%$), sensitive to medium wavelengths (such as the green color).
- S-cones ($\approx 2\%$), sensitive to short wavelengths (such as blue color).

In this way, human visual system is less sensitive to blue color than to other colors due to its amount of S-cones.

Therefore, retaining all the spectral details of the video may prove unnecessary for good quality reproduction of the video.

2.1.1.3 Temporal irrelevance

The human visual system is not sensitive to rapid changes. Thus, in the case of a video, only about 30 frames per second are necessary for humans to perceive smooth motion. Since human visual system is unable to detect rapid changes between successive frames, this can be exploited to reduce the amount of data used by a video without loss of visual quality.

2.1.2 Statistical redundancies

The statistical redundancies of each frame of a video can be used to optimize the amount of data used. Among these redundancies, we have spatial, spectral, and temporal redundancies, as in the human visual system.

2.1.2.1 Spatial irrelevance

These redundancies are introduced by a strong correlation between neighboring pixels. By the way, in an image, each pixel is correlated with its neighbors in such a way that the final result is something visible and understandable to humans. An image created with no correlation between pixels is something like a noisy image because each pixel is independent from the others. So the spatial redundancies can be exploited to predict for instance values of pixels according to neighboring pixels. In this manner, some data can be deleted, which reduces amount of data used.

2.1.2.2 Spectral irrelevance

These redundancies are created by strong correlation between neighboring pixels in color domain. This is because color changes are often gradual, and colored regions regularly extend beyond a single pixel. Thus, a pixel is strongly correlated with its neighbors in the color domain, which can be exploited to reduce amount of data used.

2.1.2.3 Temporal irrelevance

These redundancies are the most important for video compression. In fact, changes to the elements present in the video occur gradually, especially in consecutive frames. Thus, between two consecutive frames, there will not be many changes, as shown by Figure 1. Indeed, the amount of change created by this moving car is very small.

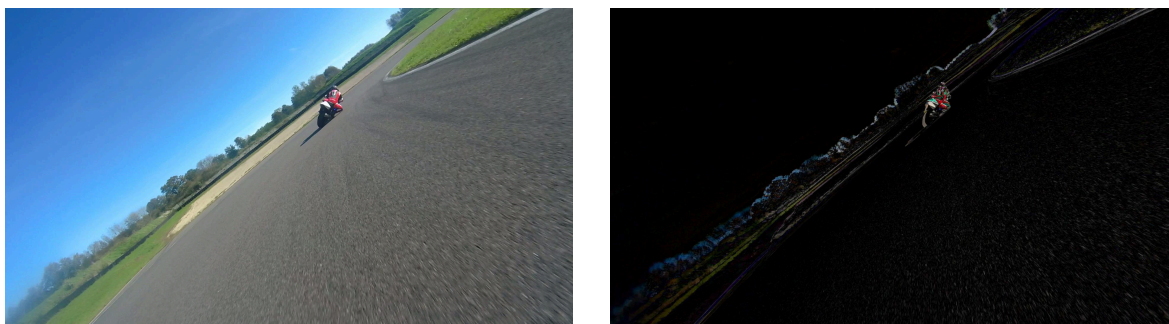


Figure 1: Difference between 2 consecutive frames for a 60 fps video

So if only the changes are recorded, most of the frame can be compressed.

2.2 H.264

The H.264 codec, also known as MPEG-4 AVC (Advanced Video Coding) or MPEG-4 Part 10, was developed in 2003. It undergoes numerous transformations and is still undergoing improvements today ¹.

2.2.1 Compression

Video compression follows steps similar to those used in image compression, which consist of data transformation, quantization for lossy compression, and data encoding for efficient

¹ Wikipedia

storage on disk. However, some additional steps are provided to exploit temporal redundancies.

More concretely, we have the steps bellow.

2.2.1.1 Color Space Transform

The frame is first transformed from RGB color space to YCbCr color space. Instead of representing the frame in RGB color space, the YCbCr color space is used to divide the frame into luminance (Y) and chrominance (both Cb and Cr).

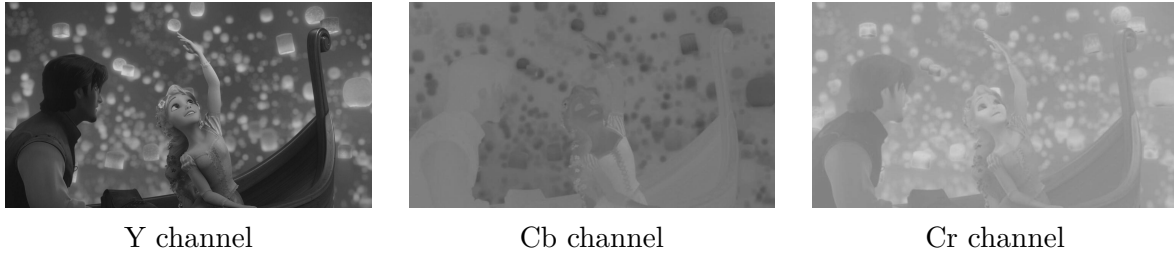


Figure 2: Image decomposed in YCbCr color space

The luminance looks after the luminosity, the brightness of the image whereas the chrominance looks after colors of the image, as shown on Figure 2. On top of that, as we can observe on Figure 2, all structural information is contained in Y channel, which represent clearly the image in grayscale.

In this way, since the human visual system is less sensitive to color than to brightness, chrominance could be compressed more than luminance at a later stage.

2.2.1.2 Split into macro-blocks

Then, the frame is divided into macro-block generally of size 16×16 . These macro-blocks perfectly represent specific regions of the image and are very useful for working on small areas of the image. This allows for easier detection of elements such as moving and not moving objects, more accurate motion prediction, and an efficient means of compressing data. Indeed, some parts of the image may be flat while others may contain a large amount of detail. And on subsequent frames, some informations can be exactly the same, as represented by the black on Figure 1. In this way, some parts of the image can be compressed more than others.

2.2.1.3 Partitioning

Each macro-block is then divided into smaller blocks depending on precision needed. This allows a better adaptation on local complexity of the frame.

The macro-block can be divided into sub-blocks of size 16×16 , 8×8 , 8×4 , 4×8 or 4×4 .

2.2.1.4 Prediction

This step is divided into two sub-steps.

2.2.1.4.1 Intra-prediction

In this case, only the current frame is taken into account. The goal is to predict the value of a pixel based on its neighborhood. The prediction mode is chosen by the encoder, such as vertical, horizontal or other modes.



- vertical mode: copy content of previous row
- horizontal mode: copy content of previous column

The prediction that gives the smallest difference between original block frame and predicted block frame is kept according to chosen mode. There exists 9 prediction modes for 4×4 blocks and 4 modes for 16×16 blocks.

The intra-prediction is very useful for scenes without moves.

2.2.1.4.2 Inter-prediction

In this case, the subsequent frames are used to make the prediction (past and future frames, depending on kind of prediction). The goal is to predict block position according to subsequent frames. This is based on the fact that only small changes are introduced between subsequent frames, as shown on Figure 1.

First of all, we need to introduce a few technical terms, summarized in the Table 1.

I-frame This is an independent frame that is fully encoded using Section 2.2.1.4.1, which results in low prediction efficiency. The file size is large due to the amount of data that needs to be retained. This type of frame corresponds to the key frames in the video, which are used to encode the other frames.

P-frame This is a frame encoded using the previous I-frame, which results in good prediction efficiency thanks to previous I-frame. The aim is to track the movement of the object in the video and use this to reduce the amount of data used. By exploiting this source of redundancy, the amount of data used is less than that of I-frame, so the compression is better.

B-frame this frame is encoded using the previous I-frame and the following P-frame, which results in high prediction efficiency. By using the previous and following images, the prediction is definitely better, allowing less data to be used than for P-frames. This kind of frame is especially used for compression.

These frames create a Group Of Picture as shown on Figure 6.

Frame type	Dependency	Prediction efficiency	File size	Goal
I-frame	No	Bad	-	Key frame
P-frame	I-frame	Good	+	Track move
B-frame	I-frame, P-frame	High	+++	Compression

Table 1: Kind of frames used in a video

There is no inter-prediction in I-frame, as described in Table 1.

On case of P-frame computation, the block of the P-frame is searched in the previous I-frame. A motion vector is then computed, describing the move of the block between this two frames. Then, the content of the reference block is predicted according to motion vector and I-frame content.

On case of B-frame computation, the block of the B-frame is searched in the previous I-frame and next P-frame. The motion vector is then computed, allowing prediction of the block according to I-frame and P-frame. In this case, the prediction of the block is more accurate thanks to the I-frame and P-frame used, allowing greater data compression later on.



Computing the motion vector is costly, however it allows to achieve great compression on moving objects for P-frames and B-frames.

The structure of the video follows a pattern as shown in the Figure 6. This pattern is called Group Of Picture (GOP) and can be used later for modification detection.

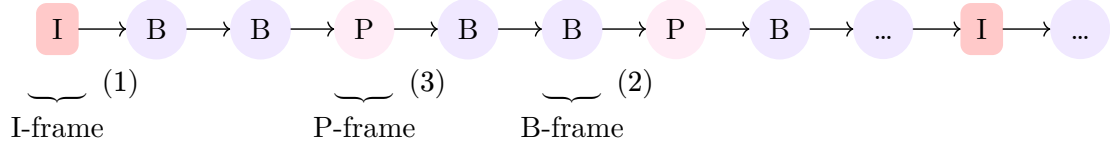


Figure 6: Frame organization in video

2.2.1.5 Residue

Once the prediction is made for a block, the difference between the predicted block and the original block of the frame is computed. Depending on results, prediction can be reapplied to have smaller residuals allowing better compression, as mentioned in Section 2.2.1.4.1.

In this way, only the method for establishing the prediction and the residual difference between the prediction and the original image are necessary to reconstruct the original image.

For instance, suppose that we have B_{original} the original block of the frame and $B_{\text{predicted}}$ the predicted block. Then, the residual R can be computed as shown in Equation 4.

$$R = B_{\text{original}} - B_{\text{predicted}} \quad (4)$$

So if only both motion vector for inter-prediction and prediction method for intra-prediction is kept with the residual, the original block can be reconstruct as shown in Equation 5.

$$B_{\text{original}} = R + B_{\text{predicted}} \quad (5)$$

This is why only residual of prediction and information necessary for prediction are retained, allowing for considerable data compression.

2.2.1.6 Transform

An integer approximation of the Discrete Cosine Transform (DCT) is applied to the prediction residual in order to convert it into frequency coefficients.

The integer approximation is used to facilitate computations and avoid rounding error introduced by a classical DCT that works with floating points. In fact, the entire approximation of the DCT is reversible and only processes integers, which is particularly relevant when working with images and videos in the integer domain.

In this way, the residual is divided into high and low frequencies, which proves useful in the next step: the quantization.

2.2.1.7 Quantization

The quantization is a lossy step that determine the quality of the compressed video. It is based on imperfection of human visual system, which is less sensitive to small details, as explained in Section 2.1.1. Since high frequencies represent small details, quantization uses the decomposition of the residue into low and high frequencies to remove the high frequencies. To do this, a quantization matrix is constructed based on the quality factor defined by the



user. Then, the residual is divided by the quantization matrix and the result is rounded to have only integer values. Some high frequencies then become 0, which reduces the amount of data to be stored. In this way, some data are lost, that's why this is a lossy step.

2.2.1.8 Entropy encoding

Finally, the quantified residues and parameters useful for prediction, such as the motion vector or prediction method, are encoded in such a way as to use as few bits as possible and to be able to reconstruct the encoded video. Depending on requirements, two methods can be used:

- Context-adaptive variable-length coding (CAVLC). This is the simplest method, but also the least efficient.
- Context-adaptive binary arithmetic coding (CABAC). This method is more complex, but allows for better compression. This is why it is preferred for high-quality compression.

Note

A very good explanation of how H264 works is available on the website abhik.xyz, which provides interactive explanations.

2.2.2 Decompression

The process of decompression follows the same steps as video compression, but in reverse order, as shown on Figure 7.

So first, data are decoded. Next, as in the quantization step for encoding, the DCT coefficients were divided by a quantization matrix, the DCT coefficients are multiplied by the quantization matrix to recover the DCT coefficients in the inverse quantization step. Then, an inverse DCT is applied to retrieve the residues, which are added to the prediction made. In this way, the original blocks of the partitioning are retrieved, as explained in Section 2.2.1.5. Then, blocks are reassembled to reconstruct the video. Finally, deblocking filters are applied to avoid blocking artifacts on the video.

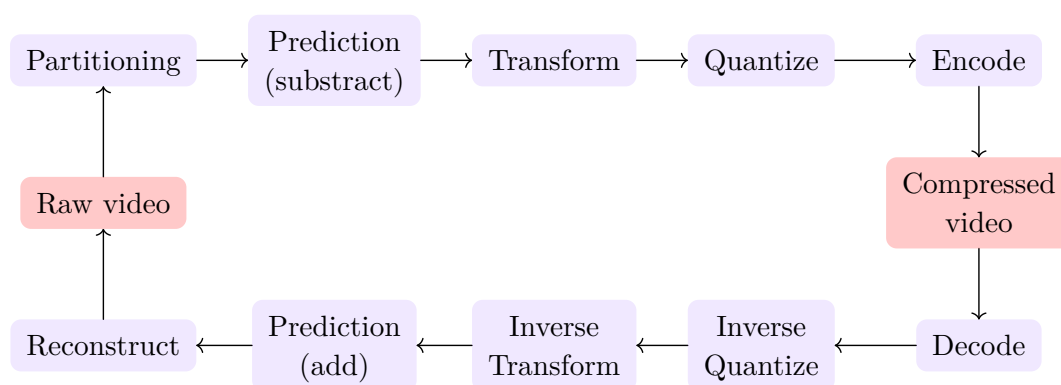


Figure 7: General scheme of video compression/decompression

2.3 H265

The H265 codec, also known as High Efficiency Video Coding (HEVC), has emerged in 2013 and is still in development. It is based on h264 and improves upon it in terms of compression



quality. The H265 codec therefore follows the compression scheme described in Figure 7, with a few modifications that make it more efficient in terms of compression.

The main difference between the H264 and H265 formats is that H264 is designed for single-core processors, while H265 is designed to run in parallel on multi-core processors. The H265 format breaks the image down into tiles that can be processed independently of each other.

2.3.1 Compression

The H265 compression follows the same steps than the H264 compression. However, both H265 encoding and decoding is made to work in parallel, allowing less power consumption. But this requires better material, making it not fully compatible with all devices.

2.3.1.1 Partitioning

This is the main step that makes H265 more efficient than H264. The partitioning structure follows a Quad Tree (QT) structure: instead of working with a decomposition into macro-blocks, the frame is decomposed in Coding Tree Units (CTU) of size 64×64 . This makes compression of flat areas such as blue skies more efficient, because instead of dividing the flat area into 16 blocks of size 16×16 , a single block of size 64×64 can be used. Thanks to the large block size, compression can be performed on higher-quality videos, up to 8K.

Then, each Coding Tree can be divided into 4 sub-coding tree, until a minimum size of 8×8 , depending on requirements. The final blocks generated by the Quad Tree structure are called Coding Unit. The figure blabla show an example of decomposition of a Coding Tree into Coding Units.

Thanks to Wavefront Parallel Processing (WWP) technology, each Coding Tree Unit can be processed in parallel, reducing computation time on multi-core processors.

2.3.1.2 Prediction

The prediction is made on Coding Units. During the prediction step, the Prediction Unit (PU) is created. It contains all information necessary to perform the prediction, such as prediction mode used or sub-decomposition of the Coding Unit.

Intra-prediction support handles 35 different modes, compared to 9 for H264, enabling more accurate prediction despite higher computational complexity. This way, the prediction is more accurate, creating a smaller prediction error that can be more compressed.

Inter-frame prediction in H265 is better than in H264 thanks to the flexibility of frame partitioning. In H264, only 16×16 blocks are used for inter-prediction, whereas in H265, block sizes can range from 8×8 to 64×64 , allowing for better motion tracking. In addition, H265 can use up to 16 previous frames to compute the motion vector prediction (with always at least an I-frame in the set of frames used). Finally, some novelties are used to increase accuracy of inter-prediction:

- Advanced motion vector prediction. This improves motion vector calculation by creating a list of candidate vectors and choosing the one that gives the best prediction.
- Merge mode. Instead of calculating the motion vector, a block can merge with a neighboring block and inherit its motion information. This is a significant improvement for uniform movements in a video, reducing calculations and the amount of information that needs to be stored.



This is why prediction is more accurate, which results in lower prediction errors that requires less data to be stored. However, prediction requires more computations compared to H264.

2.3.1.3 Transform

The transformation stage uses the prediction unit to perform a prediction. The difference between the prediction and the original block is then calculated, creating residuals that correspond to the prediction error, as in H264. Similar to H264, this prediction error is then transformed in the DCT domain and stored as a transformation unit (TU). However, compared to H264, H265 uses DCT, but also discrete sinusoidal transform (DST) to perform the decomposition of the coefficients of certain blocks of the luminance channel, which allows for better compression thanks to the structure of these specific blocks.

Finally, quantization is applied as in H264, then H265 uses an improved version of H264's CABAC entropy coding to store the results.

We can represent the final structure of H265 like in figure blabla.

2.4 H266

The H266 codec, also known as Versatile Video Coding (VVC) or MPEGi, is the successor to H265. This codec was introduced in 2020 and is still under development. It enables compression of 360-degree and high dynamic range (HDR) videos, supports video compression up to 16K quality, and brings improvements to video compression, making it up to twice as efficient as H265.

2.4.1 Compression

2.4.1.1 Partitioning

The main change bring by H266 is its partitioning structure more flexible than the Quad Tree structure of H265: the Multi-Type Tree (MTT) which is a mix-up between Quad Tree, Binary Tree and Ternary Tree. The partition unit is still called a Coding Tree Unit, but can reach a size of 128×128 , which is more efficient for high-resolution videos such as 8K or 16K videos.

Each time the frame is divided into a coding sub-tree, the division can be performed as follows:

- Binary tree: 1 cut of the area, horizontal or vertical
- Ternary tree: 2 cuts of the area, horizontal or vertical
- Quadratic tree: area divided into 4 quadrants.

The new partitioning system therefore allows for more accurate tracking of object shapes.

2.4.1.2 Prediction

Instead of having 35 different prediction modes, H266 introduces 67, allowing for greater accuracy during prediction.

In addition, some new tools are introduced for intra-prediction, such as:

- Matrix-based Intra Prediction (MIP). This tool uses a prediction matrix that has been predefined through prior learning.



- Intra Sub-Partition (ISP). The block can be broken down into 2 or 4 sub-blocks, each of which will have its own prediction.
- Cross-Component Linear Model (CCLM). Chrominance prediction is improved by using the existing correlation between chrominance and luminance.

And for inter-prediction:

- Affine Motion Compensation, allowing affine transformation of the block, such as zooming, rotation, etc. This is extremely useful for objects rotating or zooming.
- Decoder-side Motion Vector Refinement (DMVR), a new feature added to the decoder side that allows the decoder to refine the received motion vector. This means that the transmitted motion vector can be of lower quality.
- Bi-directional Optical Flow (BDOF). This type of tool is generally implemented in video editing software such as DaVinci Resolve to avoid jerky slow motion. The principle consists of increasing the prediction based on future and past images. In software, this method is used to add additional images predicted from previous and past images in order to smooth out slow motion.
- Geometric partitioning, which allows the block to be divided into two sub-blocks (for example, by making a diagonal cut) and different movements to be applied to each sub-block. This makes movement tracking more accurate.

2.4.1.3 Transform

For transformation, since H266 doesn't partition images into blocks, the transformation can also be applied to non-square blocks. The encoder can choose between different transformations, such as Discrete Cosine Transform VIII and Discrete Sinus Transform VII, in order to apply the best transformation for a data compression.

2.4.1.4 Quantization

Unlike the H265 or H264 codec, the H266 codec uses adaptive quantization, called Dependent Quantization. During the rounding phase of quantization, the coefficient value will be rounded based on the previous coefficient, which will increase the efficiency of entropy coding.

For instance, if we have the sequence $[0.2, 0.3, 0.6]$, normal rounding will return $[0, 0, 1]$. However, dependent quantization will return $[0, 0, 0]$, because for 0.6, it will detect that the previous value was 0, so instead of putting 1, it will put 0, which does not break the existing sequence of 0. The sequence 0 is therefore preserved and increased, which improves the entropic coding of the sequence.

2.5 AOMedia Video 1 (AV1)

The AOMedia Video 1 (AV1) codec is a royalty-free and open-source codec that has been developed by "Alliance for Open Media", an alliance between Amazon, Google, Netflix, VideoLAN and other actors. The first version was released in 2018 and this codec is still in improvement. This codec was created to success the VP9 Google codec for streaming video on internet and social network and is now used and supported by some range of devices.

The base principle stay the same. However, the method differs from others. First of all, the AV1 remove the video noise, avoiding complex encoding. Then, during decoding, some



noise is artificially created and added to the image, allowing better stream speed with approximatively same result...

2.5.1 Partitioning

The frame is decomposed in SuperBlocks (SB). At the beginning, the encoder choose between SuperBlocks of size 128×128 or 64×64 . Then, each block can be divided in 10 different way:

- none: the block is not partitioned
- split: partitioning into 4 square sub-blocks
- horizontal: horizontal division of the block
- horizontal 4: the block is divided into 4 equal horizontal strips
- vertical: vertical partitioning of the block
- vertical 4: the block is divided into 4 equal vertical strips
- horizontal A: the block is first divided horizontally, then the upper part is partitioned vertically
- horizontal B: identical to horizontal A, but with the lower part partitioned vertically
- vertical A: the block is first divided vertically, then the left part is partitioned horizontally
- vertical B: identical to vertical A, but with the right sub-block partitioned horizontally

The minimal block size is of 4×4 . Thanks to this 10 different way of partitioning the blocks, the AV1 codec is able to effectively detect the shape of complex objects. In addition, thanks to the size of SuperBlocks, high-resolution videos with flat areas such as blue skies are processed efficiently.

Compared to H266 partitioning, the AV1 codec performs less complex partitioning, but thanks to predefined partitioning shapes, partitioning can be performed more efficiently than with H266, and it is easily parallelizable with hardware acceleration already implemented in many devices, such as NVIDIA graphics cards since the 3000 series.

2.5.2 Prediction

The AV1 codec uses up to 56 prediction modes. However, some new tools are used to achieve an accurate prediction, such as for intra-predictions:

- Paeth Predictor: this is an algorithm taken from PNG format that search best prediction according to up, down and diagonal pixels.
- Smooth Predictor: tool optimized for progressive gradients
- Chroma from Luma (CFL): a tool that predicts chrominance from the luminance channel. Since the luminance channel offers better quality, this allows for accurate and high-quality painting of the chrominance channel. This tool is one of the best data savers in AV1.

And for inter-prediction:

- Overlapped Block Motion Compensation (OBMC): this tool consists to take motion vectors of neighboring blocks and mixing them into block's border, allowing smooth block transition.
- Wrapped Motion: tool that looks after small affine transformations such as zoom, rotation, etc...
- Compound Prediction: tool that combine multiple reference frame with complex masking. The idea is that certain objects can be better predicted from previous images, for example, and that other objects in the same video can be better predicted from subsequent images.



In this way, a mask is created to retain the best prediction for the object and the other object based on the previous and subsequent images.

As with H266, the prediction obtained using the new tools and methods is more accurate, but at the cost of longer computation times.

2.5.3 Transform

AV1 can achieve transformation on blocks of size up to 64×64 . As a 2D transformation can be applied first in 1 dimension and then on the other dimension, the AV1 use a combinaison of different kind of transformations.

- Classical DCT transformation
- Asymmetric Discrete Sine Transform (ADST) used for directional gradients
- FlipADST that is only ADST applied in inverse order (right to left or down to up). In this way, directional gradients are well managed.
- Identity (IDTX): no transformation. This is very usefull for brutal transitions like black text on white, avoiding some artifacts introduced by a transformation.

Thanks to this 16 combinations of transformations, information is more preserved in quantization step.

2.5.4 Quantization

The quantization used in AV1 is a granular quantization named Delta-Q, allowing dynamic quantization. On top of that, the quantization is less visible for human visual system.

2.5.5 Entropy encoding

Use multi-symbols arithmetic coding that uses symbols instead of bits. This allows a better parallelilization on modern Computing Units.

2.5.6 Decoding

The decoding use:

- Deblocking filter
- Constrained Directional Enhancement Filter (CDEF) that detect edges of blocks and remove noise across edges without destroying it...
- Loop Restoration that restore global quality of a block using denoising filters such as Wiener filter...

3 Forensic

Now that we know how video compression works for different encodings, we will look at how digital forensics can detect certain modifications to a video.

To do this, we will review the artifacts generated at each stage of video compression and see how they can be exploited to detect certain problems in a video.

There is different kind of video forgery detections, as shown on Figure 8.

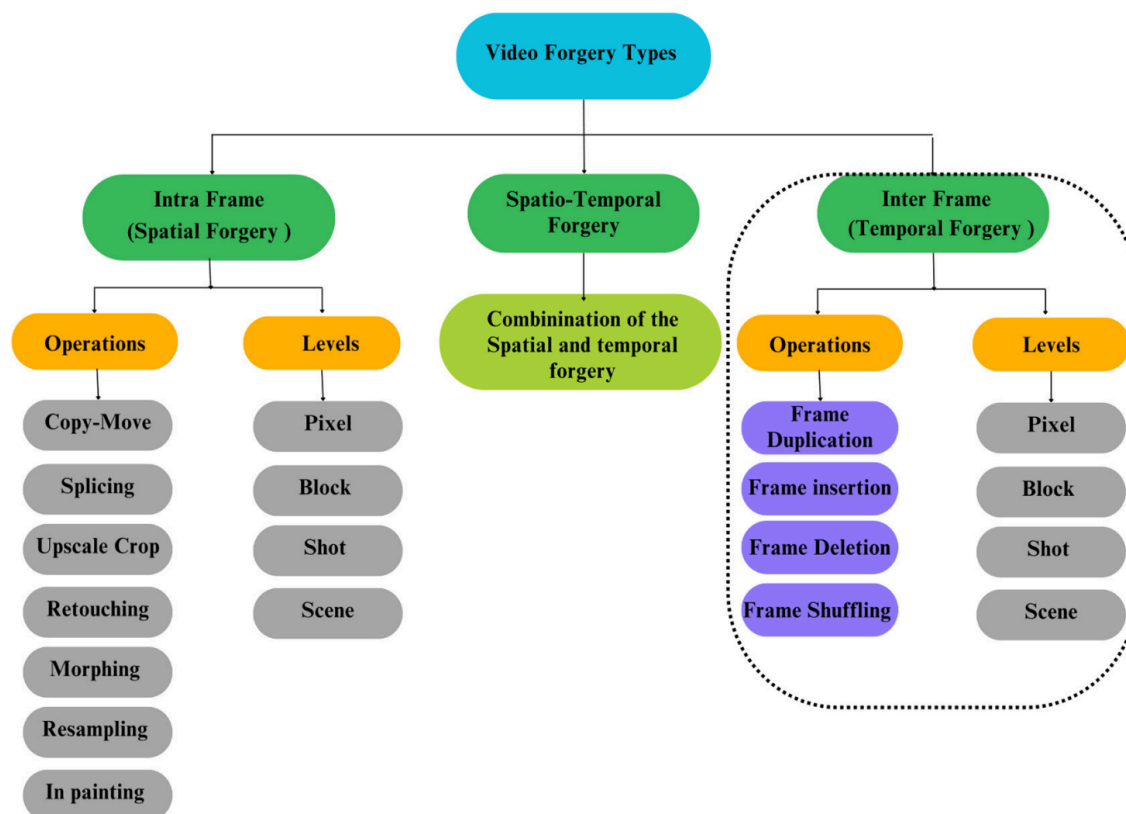


Figure 8: Video forgery classification [1]

3.1 H264

3.1.1 Stream Structure

As shown on Figure 6, the H264 like the other codecs follow a specific GOP structure. The modification of this structure can lead in some irregularity that can be detected by the GOP structure analysis, as shown on Figure 9.

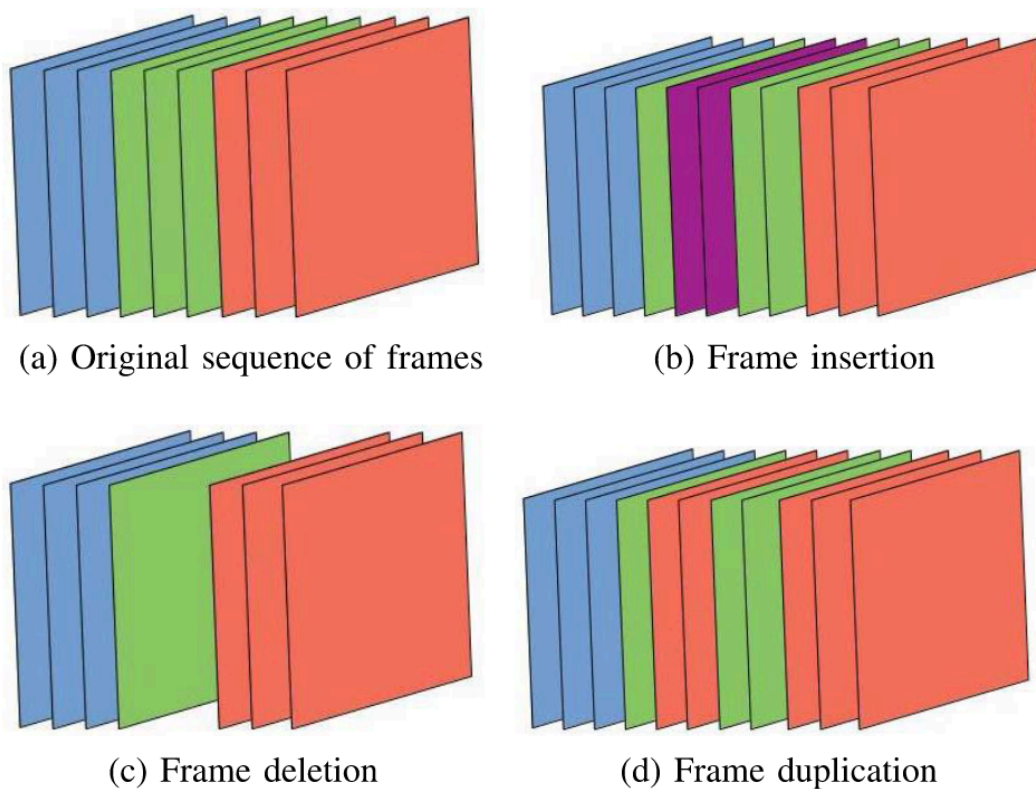


Figure 9: Modification of GOP structure [2]

3.1.2 Double Compression

3.1.3 Intra-frame manipulation



4 Results



5 Discussion



6 Conclusion



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